

HAO HU

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SUMMARY

Staff-level software engineer with 10+ years at Google, Uber, and Microsoft building large-scale distributed systems, data infrastructure, and platforms — now applying that foundation to production AI/agent systems as a startup CTO.

Production AI / agentic systems builder — RAG, AI data analytics, and tool-using agents with deterministic guardrails, citation-grounded refusal, human approval for risky actions, and golden-set regression evals.

Hands-on engineering leader with strong CS fundamentals and architecture judgment — leads small teams from architecture through delivery, using spec-driven AI-assisted workflows to ship faster without sacrificing reliability.

TECHNICAL SKILLS

LLM & Agents: agent orchestration, tool / function calling, MCP integration, workflow/state graphs, multi-agent routing, context engineering / memory, LangChain / LangGraph / DeepAgents, OpenAI / Claude / DeepSeek

AI-Native Engineering: spec-driven development, AI-assisted implementation, TDD / validation, review-gated output, harness engineering, Claude Code / Codex / Cursor / OpenCode

RAG & Retrieval: RAG (multimodal), hybrid retrieval, embeddings, reranking, query optimization, Milvus, Qdrant

Evaluation & Safety: eval harnesses, golden-set regression, RAGAS, LLM-as-judge, guardrails, human-in-the-loop

Inference & ML: PyTorch, Transformers, vLLM, SGLang, Ollama, Hugging Face, XGBoost / LightGBM / CatBoost

Programming Languages: Java, Go, Python, SQL, C++, C#

Backend & Distributed Systems: microservices, API design, gRPC / Protobuf, REST, Thrift, Spring AI, FastAPI, SSE

Data Systems: data modeling, CDC, PostgreSQL, MySQL, Redis, BigQuery, Kafka, Flink, ClickHouse, MongoDB

Cloud & Platform: Google Cloud (Cloud SQL, GCE, GCS), Docker, Kubernetes, CI/CD, observability

EXPERIENCE

Yuanbao Creative Technology

Feb 2025 – Present

Chief Technology Officer

Wuhan, China

- **Architected and shipped a unified AI layer for an ERP SaaS platform** across operations, analytics, and knowledge; established shared eval/safety practices: golden sets, regression suites, least-privilege tools, and approval gates.
- **Operations Copilot:** built a **DeepAgents**-based agent harness over ERP entities (product / inventory / order / supplier / PO), using role-scoped **MCP** tools, specialist delegation, sandbox-isolated execution, auditable tool traces, and a **Spring AI / Java 21** server with human-in-the-loop approval for risky writes.
- **Analytics Agent:** delivered self-service NL-to-SQL over the ERP **ClickHouse** warehouse, backed by a **Qdrant** semantic layer that cut prompt context **73.2%** via focused retrieval; enforced deterministic **SQLGlue** guardrails, scoped read-only execution, bounded SQL repair, and result-equivalence / safety regression evals.
- **Knowledge Assistant:** delivered **LangGraph**-orchestrated enterprise-document RAG with multi-source ingestion and **Milvus** hybrid retrieval; supported strategy-aware retrieval (precise / balanced / broad), query expansion, HyDE, RRF/rerank, and a retrieval test bench while enforcing citation grounding and strict-evidence refusal.

Uber

May 2022 – Mar 2023

Senior Software Engineer

Toronto, ON

- **Built Uber's first CPG (consumer-packaged-goods) Ads feature — storefront banner ads:** led backend delivery on a 3-person team, building a Go/gRPC/Thrift backbone service backed by DocStore between the UberEats storefront and Uber Ads so advertisers like Coca-Cola and Red Bull could run item/brand campaigns across retail chains; **scaled to ~250K QPS at ~140 ms p99** in its first quarter.
- **Redesigned the CPG ad-campaign data model:** replaced restaurant-era single-location item IDs with a GTIN-keyed model bundling brand, city, and country, enabling national campaigns while eliminating millions of duplicate campaign entries and manual sales-ops burden.
- **Launched store-menu sponsored items** by extending the CPG Ads backbone into a third-party ad integration: ranked sponsored items in storefront sub-menus and streamed impression/click events through a Kafka + Flink billing pipeline; **held ~180 ms p99** with the external provider in the request path, backed by Prometheus / Grafana monitoring and dependency alerting.
- **Drove the staged production rollout of CPG Ads** and served as the Ads team's dev representative — aligning data science, sales, DevOps, partner teams, and the external ad provider.

Google

Feb 2018 – Feb 2021

Software Engineer — Cloud SQL (Data Integration / Disaster Recovery)

Sunnyvale, CA

- **Shipped Cloud SQL PostgreSQL CSV import / export to GA:** hardened the injection-sensitive import / export path with on-demand, least-privilege database access and Cloud SQL IAM for Google Cloud Storage flows.
- **Built customizable CSV formatting across MySQL and PostgreSQL:** extended the backend REST API and redesigned import / export one-shot tasks in Go, enabling reliable export / import round-trips.
- **Raised backup health from 99.992% to 99.997%** across a large fleet backing a 99.95% SLA: owned and hardened the backup-verification pipeline (snapshot attach, DB boot, checksum on pooled GCE instances), resolving top failure classes including resource exhaustion, startup failures, and false-positive verification failures.
- **Enabled PITR-safe transaction-log retention:** built a log-cleanup system for backup snapshots that preserved recovery-critical MySQL logs while pruning the rest to reduce fleet-wide snapshot storage, allowing instances to restore to any timestamp via nearest backup + log replay.
- **Kept managed instances recoverable:** replaced a fixed-count retention window that pruned oldest-first regardless of health with a policy that always retained the most recent backup marked healthy by the verification pipeline.
- **Supported Cloud SQL production reliability:** served on-call across the Java backend and Go on-instance agents, resolving release-blocking and customer-escalated issues, and mentored junior engineers.

Microsoft

Jan 2014 – Oct 2017

Software Development Engineer

Mountain View, CA

- **Independently designed and built a distributed tool to profile 400M+ legacy Hotmail accounts (C++),** collecting account characteristics so the Outlook migration's account selector could stage cutovers and close 40+ feature gaps; partnered with the migration team through a successful migration.
- **Co-built "People I Communicate With" (PICW) from the ground up** on a 3-person team — an Outlook component (C#) that efficiently captures and stores user communication signals (frequent contacts; mail read-and-ignored, left unread, junked, or reported as spam/phishing), initially powering Contact Safety and anti-spam.
- **Took sole ownership of PICW and scaled it,** feeding signals into the People Relevance people-graph and serving as dev representative across teams; enabled by default in Outlook on the web, IMAP, and POP for enterprise and multi-tenant customers (100M+ Office 365 commercial users in 2017).
- **Served on the on-call rotation,** resolving release-blocking and customer-escalated issues.

Epic Systems

Jun 2012 – Nov 2013

Software Developer

Madison, WI

- **Built and maintained EMC2,** Epic's internal development-management platform — artifact tracking, source control, and deployment monitoring (an early CI/CD system) — in VB.NET on InterSystems Caché.
- **Developed customer release and upgrade utilities** (console and web) for software upgrades and version checks, streamlining how customers kept their deployments current.
- **Provided developer enablement and escalation support,** unblocking development and deployment issues and delivering a Trial Dev Environment service built on a multi-way merge for safe change validation.

HCR ManorCare

Jan 2011 – Apr 2012

Data Warehouse Developer

Toledo, OH

- **Built SSIS ETL pipelines** loading multi-source data into an enterprise star-schema data warehouse, with validation, logging, and error handling.
- **Developed advanced SSRS reports** (cross-tab, drill-down, parametric, cascaded) and T-SQL stored procedures delivering finance and operations business intelligence.

EDUCATION

Bowling Green State University

M.S. in Computer Science

Aug 2010 – May 2012

GPA 4.0/4.0 · Bowling Green, OH

Wuhan University

B.S. in Software Engineering

Aug 2006 – Jun 2010

GPA 3.5/4.0 · Wuhan, China